When Schema Lexicons Fail: Diagnosing and Repairing Entity-Layer Collapse in a Production Knowledge Base

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Abstract

Background and problem. In dictionary-driven entity enrichment, the schema lexicon effectively serves as the extraction model. We report a production incident in which a category-word lexicon silently collapsed the entity layer of a 61,137-document scholarly knowledge base: 575 type-level entities ("method", "energy") carrying ∼19.2M links while every job-level monitor stayed green. **Method.** We diagnose the failure across three layers (extraction mechanism, lexicon provenance, monitoring blindness), repair it with a three-source corpus-presence-attested lexicon (3,702 terms), rebuild the entity layer under production load, and derive gold-free quality monitors led by a link-concentration diagnostic invariant to uniform rescaling of link counts. **Results.** An isolated fixed-code, fixed-corpus experiment that varies only the lexicon reproduces the collapse signature alone (109±5 entities, top-10 concentration 0.513±0.016, 1.85±0.05 concepts per matched document) and isolates the repair effect (692±27, 0.300±0.008, 3.89±0.07; mean±sd over ten random resamples of the same corpus); the rebuilt layer carries 1,871 entities, 25.1M links, and a functioning relation layer; distinct concepts per document rise 15.7→37.8 with annotated-document coverage ∼51.5%→98.5% (the pre-repair snapshot had 48.5% of documents with no annotations); top-10 link concentration falls 0.419→0.345; a risk-stratified 300-link audit (187 distinct entities; a stratum-mixed rate, not a link-uniform estimate) reads 0.84 overall—0.95 on domain terms, 0.58 on short ambiguous symbols (a second Qwen-series model agrees on 96% of its parseable subset). **Limitations.** The generic tail survives repair (9% of links sit on the single

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ambiguous symbol "be"); retrieval-facing gains are not attributable to the lexicon alone (entity-channel ablation is @10 metrics unchanged; alias expansion +1.5pt, ns); the isolated-experiment metrics are structural rather than semantic; audit and cross-judge are model-assisted, not human-adjudicated; the concentration alarm bands are candidate thresholds read post-hoc from 13 libraries.

Keywords: knowledge base construction, schema quality, terminology extraction, entity linking, production incident, observability

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1. Introduction

Large scholarly knowledge platforms increasingly generate their subject schemas with LLM assistance: a batch tool drafts entity types with a handful of aliases each, a validator checks structural well-formedness, and the schema is bound to knowledge bases as configuration (our own batch generator is the primary failure source in this account; LLM-for-ontology generation is surveyed in [5]). This is attractive at scale—our deployment covers 80,855 parsed journal articles (80,852 active, 7,229,225 text chunks) across two large knowledge bases and several smaller ones—and it keeps humans out of a loop that would otherwise require domain lexicographers per discipline.

This paper is about what that design decision does to the entity layer when the enrichment pipeline extracts mentions by dictionary scan. In our system, stage-K1 mention extraction is deliberately LLM-free: a schema's `entity_types` main names and aliases constitute the scan dictionary. The choice is defensible—dictionary scan is cheap enough to run over 7.2 million chunks, deterministic, idempotent under re-ingest—and it performed adequately on the knowledge base whose schema had been hand-evolved over years. It failed catastrophically, and silently, on the two knowledge bases whose schemas came out of the batch generator.

The failure signature: a 61,137-document nuclear-electronics knowledge base (104112E6) accumulated **19.2 million entity links onto 575 canonical entities**; the entities were not domain concepts but type words—"method", "parameter", "energy". A 13,408-document physics knowledge base (693372D5) showed the same failure pattern at 2,018 entities. Meanwhile every operational signal was green: enrichment jobs completed normally, link volume read as growth, and retrieval endpoints answered. The entity graph had degenerated into a bipartite "type-to-

document" index while presenting, to every surface-level check, as a healthy knowledge graph.

We contribute:

1. **A failure mode and its causal confirmation.** A forensic record of entity collapse induced by schema-lexicon quality (Section 3), elevated from correlation to cause by a controlled lexicon-only experiment (Section 6): with pipeline code and corpus both fixed, swapping the dictionary alone reproduces the collapse signature and its repair across ten random resamples of one corpus. The observables that expose the collapse are the scale-invariant concentration of links on the top entities and the top-entity name distribution, with entity cardinality as a corroborating (and manipulable-in-both-directions) secondary signal.
2. **A corpus-attested lexicon lifecycle, deployed through a full-chain rebuild at production scale** (Sections 4–5): 2,284 transplanted seed terms from a healthy near-domain library, 7,391 corpus-frequency candidates classified by an LLM, and bounded per-type expansion—yielding 3,702 real terms for nuclear electronics (v2.1) and 1,279 for physics (v1.1), 223 legacy conflicts repaired in passing—followed by vertex-and-table-scoped teardown and 766,501 re-enqueued jobs, with the pipeline pathologies the rebuild surfaced, each with a measured fix (most dramatically a 137× link-throughput recovery once an $O(N^2)$ alias check, amplified by the richer lexicon itself, was cached).
3. **Gold-free entity-quality observability** (Sections 6–7): before/after evidence (cardinality 575→1,871 with a functioning relation layer; scoped retrieval trajectory +6.6/+7.3 cov@10 with a channel-level attribution bound; a 300-link stratified audit), the four monitors—led by top-10 link concentration—calibrated on 13 production libraries, and a corrected incident log showing that dashboard-derived incident narratives themselves require row-level verification.

The larger claim we defend: in dictionary-driven pipelines, **the schema lexicon is not descriptive metadata but operative extraction surface**. Treating it as configuration-file decoration—easily batch-generated, rarely audited—is a structural mistake that this incident renders measurable.

2. Background: dictionary-driven enrichment

Our enrichment pipeline (four stages per document: mentions → links → relations → graph) extracts mentions in stage K1 by normalized dictionary scan: every schema alias match over chunk text yields a `raw_mention` with an entity-type hint; stage K2 links mentions to canonical entities by normalized form + type; K3 lifts selected mention pairs into typed relations with an LLM pass; K4 projects the per-document entity subgraph into the graph store and search-side entity index. Retrieval consumes all four: an entity channel seeds personalized PageRank, a co-occurrence channel, and entity-vector similarity all trace back to linked mentions.

Why no LLM in K1? Cost and determinism. At 7.2M chunks, a per-chunk LLM extraction pass is a material recurring expense and a re-ingest hazard (non-determinism breaks idempotent rebuild); a scan is $O(\text{corpus})$ with zero marginal model cost. The design bets everything on one input: the dictionary. That bet is invisible in the architecture diagram and total in its consequences—a point Section 3 makes empirical.

The healthy control in our own deployment: a hydrogeology knowledge base whose schema was hand-evolved to 3,742 genuine domain terms (78 documents → 1,608 canonical entities, well-distributed). The contrast cases: schemas produced by our batch tool, whose output per entity type is a small set of *category* aliases (4–8 per type)—a taxonomy of the subject, not a terminology of it (the distinction between terminological and taxonomic vocabulary is classical [6], and the adjacent question of how to measure knowledge-graph quality is itself an established literature [3]).

Neither half of our pipeline is novel in isolation. Dictionary- or gazetteer-driven entity recognition is one of the oldest NER traditions [2], and corpus-statistical term recognition with linguistic evidence goes back to C-value/NC-value [1]—our corpus-candidate classification source (Section 4) is that automatic-term-recognition program reborn with an LLM in the classifier seat. Likewise, detecting and repairing degraded knowledge-graph content is an established concern [3], and production data-validation systems routinely gate model inputs with schema- and statistics-level checks [4]. What this incident supplies is a specific failure locus: when the dictionary *is* the extraction model, a structurally valid but terminologically hollow schema passed every gate *this pipeline had in place*—schema-shape validation, ingestion checks, job-level monitoring. That is a statement about our gates, not about the prior art: distribution-statistics monitoring

Table 1: The collapse, measured. "bckp." = 2026-10-06 cold-backup snapshot (mid-enrichment); "disc." = discovery-day counts; the hydraulic reference case is measured 2026-10-10.

KB	Docs	Entities	Links	Schema provenance
104112E6 (nuclear)	61,137	**575** disc. / 503 bckp.	∼**19.2M** disc. / 4.81M bckp.	batch v2.0
693372D5 (physics)	13,408	**2,018**	2.23M (bckp.)	batch
hydrogeology (ref.)	78	1,608	719,515	hand-evolved

is the right tradition, but the statistic must be aimed at the lexicon's terminological content—none of our gates asked whether a lexicon's entries are domain-specific terms, the question term-recognition research takes as its starting assumption. The monitors we propose (Section 7) target that acceptance gap.

3. The collapse, forensically

The observation. During an unrelated storage investigation (2026-10-08), a routine entity-quality sample on 104112E6 returned type words where domain concepts were expected. The full picture (Table 1):

61,137 documents carrying ∼19.2M links is ∼315 links per document on average, and ∼33,000 links per entity: every document "mentions" the type vocabulary constantly. The graph is a bipartite type–document index that is row-level indistinguishable from a populated knowledge graph.

The incident timeline (dates stated plainly, because a repair of this scale compressed into a two-day window is exactly the kind of claim a reader should be invited to check):

The compression is real and we do not smooth it over: the tooling was written under incident pressure on day one, and the evaluation experiments are retrospective audits of a settled production state, not a pre-registered protocol—a limitation Section 9 keeps visible.

Layer 1 — mechanism. K1 scans for schema aliases. When the alias set for type `method` contains common words ("method", "approach", and their CJK equivalents), every occurrence of those words in 7.2M chunks be-

Table 2: Incident timeline (events compressed; full narrative in the text).

Date	Phase	Event
10-08	discover	storage investigation surfaces type-word entities
10-08	root-cause	three-layer diagnosis; lexicon tool built, run, committed (v2.1)
10-08	rebuild	teardown + 766,501-event re-enqueue; B95 silent misattribution window
10-09	drain	four stages complete; stats recomputed; vectors reprojected
10-09	extend	two smaller libraries with the same failure pattern treated (same playbook)
10-09/10	audit	scoped refresh, channel ablation, link audit, concentration fleet-wide
10-09/10	manuscript	drafted, revised; retrospective — not pre-registered

category-word lexicon (v2.0)

`method` → type *method*
`energy` → type *energy*
`time` → *physical_quantity*

575 entities / 19.2M links
48.5% of documents unannotated

repaired lexicon (v2.1)

`hpge detector` → *detector*
`preamplifier` → *circuit*
`neutron irradiation` → *isotope*

1,871 entities / 25.1M links
98.5% of documents annotated

Figure 1: Mechanism: the schema lexicon *is* the extraction model. Identical scan-and-link code over the same corpus; only the dictionary differs (terms and examples illustrative of each lexicon's content).

comes a mention, and K2's normalized-form+type linkage folds them onto one canonical entity per (form, type) pair. The mechanism executed faithfully on degenerate input.

Layer 2 — provenance. The dictionaries were the schemas. The batch generator, prompted to "draft entity types with aliases for subject X", produced exactly that: type-level category vocabulary. Nothing in its validator checks whether aliases are *domain-specific*—structural validity is the gate, and category words are structurally valid.

Layer 3 — the reference case. Hydrogeology's 3,742 hand-evolved terms over 78 documents yield 1,608 entities. We call it a reference case, not a controlled comparison: domain, scale, register, and lexicon provenance

all differ, so it establishes plausibility, not causation — which is what the isolated experiment of Section 6 (6.5) is for.

Why nothing fired. Job-level observability was all green (jobs DONE, throughput nominal, error rate zero). Link volume—if anything—read as *success* (19.2M links!). The observables that expose the collapse are (a) the concentration of links on the top entities—the collapsed state measures 0.419 with its top-10 entities being pure type vocabulary, versus 0.18 for the healthy control (Section 7's monitor (a) gives the measurement and its provenance)—and (b) the name distribution of top-linked entities, which is pure type vocabulary. Entity cardinality per document—a factor of 2,192 below the reference case (0.0094 vs 20.6), over three orders of magnitude—corroborates, but cardinality is a coarse signal gameable in both directions (the small-library treatment of Section 9 shows category-word aliases inflating it), which is why it is the secondary, not the primary, monitor. Neither observable is provided by standard pipeline dashboards by default. This blindness generalizes to any dictionary-driven system whose dashboards count jobs and rows.

4. Corpus-Attested Lexicon Construction and Validation

Figure 2 shows the three-source admission process and its quality boundary.

The repair requirement: produce genuine domain terminology at scale, without per-subject lexicographers, without hallucinated terms—and make it a repeatable tool, not a one-off patch. Our generator implements three sources, all gated:

1. **Seed transplant (2,284 terms).** A healthy near-domain library (the RADAI corpus knowledge base, same academic register) donates its terms with types. Transplant is the highest-precision source; corpus attestation below still applies.
2. **Corpus-candidate classification (7,391 candidates).** High-frequency n-grams from the target corpus's own text are proposed—concretely, the top 8,000 n-grams (unigrams and bigrams) by chunk frequency over a 3,000-chunk-per-KB sample, keeping candidates with frequency ≥ 5 after a boilerplate and stop-word filter (section vocabulary such as *acknowledgements/approach/using*; closed-class English words). An LLM classifies each surviving candidate into a schema entity type or rejects it. The corpus floor is a corpus-presence

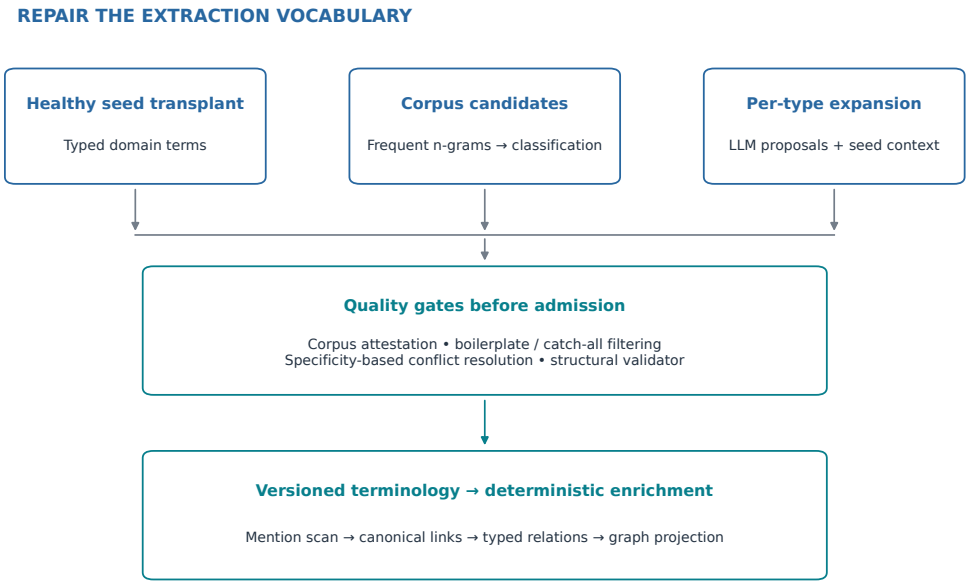


Figure 2: Three-source terminology generation and deterministic enrichment. Healthy seeds, classified corpus candidates and per-type expansion proposals pass corpus-attestation, vocabulary, conflict-resolution and structural gates before entering a versioned schema. The admitted vocabulary drives mention scanning, canonical linking, typed-relation adjudication and graph projection. Attestation verifies occurrence; contextual sense and domain specificity still require an independent link-quality audit.

gate—attestation means the term occurs in the corpus, which is necessary but not sufficient for termhood (the "be" case of the name-quality section is exactly this gap); it prevents hallucinated entries, not generic-word admissions.

3. **Per-type expansion ($\leq$60 per type).** For each entity type the LLM proposes additional domain terms, given the type description and the transplanted seeds as in-context anchors; the same corpus-attestation gate applies. The bound keeps any single type from flooding.

Generation parameters, for reproducibility: the classifier/expander is an AWQ 4-bit build of the Qwen3.6-35B-A3B checkpoint (community build `cyankiwi/Qwen3.6-35B-A3B-AWQ-4bit`; we retain the deployment string verbatim since the endpoint serves it under a historical alias), served

via vLLM behind a round-robin router, temperature 0.2, ≤4,000 output tokens per call, three retries on empty or non-JSON responses; the entire three-source run for both schemas is a single idempotent invocation of the generator.

Quality gates beyond attestation: *catch-all types* (`term`, `term_cn`, `model_or_code`, `concept`, `question`) accept no classified noise—they are the natural sink of classification uncertainty and the main entry route for taxonomy garbage; a boilerplate filter removes layout/section vocabulary; cross-type alias conflicts resolve by a specificity ordering (the more specific type wins); and the schema validator runs as a final structural gate before any version bump. The same pass repaired 223 legacy cross-type conflicts in existing schemas.

Output: **nuclear_electronics v2.1 = 3,702 terms; physics v1.1 = 1,279 terms**. 104112E6 rebinds to nuclear_electronics (its corpus is nuclear-domain journals). Both versions are idempotent re-runnable—the tool is now part of the schema lifecycle, not an incident artifact.

An honest note on the asymmetry (physics 1,279 versus nuclear 3,702): the physics corpus is smaller and its seed transplant thinner; v1.1 is a floor, not a ceiling, and the runbook treats small-physics libraries as follow-up work (Section 9).

5. Rebuild at scale

Repairing live knowledge bases requires tearing down the collapsed entity layer without stopping ingestion or retrieval. The rebuild driver executes per knowledge base: delete graph-store vertices with edges; delete the four relational tables (mentions, links, relations, graph-projection queue rows) in bounded chunks; delete the search-side entity projection by query; then re-enqueue all documents through the four stages—**766,501 enqueue events** across the two affected libraries. The count is events, not distinct doc-stages (74,545 documents × 4 stages = 298,180): the surplus is the repair window's re-enqueue waves—the silent-empty links reset (75,831), the post-fix relations/graph zero-effect resets (≈149k), the watchdog-deletion graph recovery (≈122k), the small-library treatment (5,772), and re-dispatch of partially processed waves during the gated window; the job table holds one row per doc-stage with an attempts counter, so every wave re-uses rows. The rebuild ran against the production fleet (five multi-worker enrich instances, one links-dedicated consumer per KB) while normal retrieval served.

The rebuild doubled as the system's first full-corpus stress test of the *rich-lexicon* configuration, and four pathologies surfaced—each now a fixed, committed, measured lesson:

- **B90 — unbounded projection join.** Graph projection originally joined each document against *whole-corpus* entities and the *whole-corpus* link table; at 80k documents and millions of links a single projection took 25+ minutes, and 80 concurrent consumers saturated a 112-core MySQL (107 cores, load 500). Scoping the join to the document's own mentions (an index-ordered path) is semantically equivalent—an entity projects through documents that mention it—and took projection to seconds, database to 29 cores.

- **B91/B93 — link-insertion deadlocks.** Two deadlock classes in link insertion: unordered INSERT-SELECT acquiring gap locks in interleaved order (fixed by pinning insertion order to `mention_id`), and per-action round-trip inserts (fixed by folding all actions into a single UNION-ALL derived-table INSERT-SELECT). Deadlock error 1213 went to zero.

- **B94 — the $O(N^2)$ the new lexicon amplified.** Alias canonicalization checked every acronym pair quadratic in alias count—95 of 101 seconds per links task, burning the GIL while 16 threads queued. Caching canonical forms by schema signature (lexicon version + alias count) took it to 0.05 s. Net link throughput: **8/min → 1,100/min (137×)**—the collapse-era lexicon was so small it hid this bug for months; the *repair* exposed it in an afternoon. The 137× is a window-level aggregate over concurrent fixes (the deadlock repairs land in the same window); the cache's isolated contribution is measured at task level—95 of 101 seconds per links task—and the cache key (schema version + alias count) can in principle alias two same-size, same-version, different-content lexicon edits; version-pinned bumps (v2.0→v2.1) are the mitigation.

The last point deserves emphasis in prose, not just as a bullet: the collapsed lexicon (575 de-facto terms) masked an $O(N^2)$ hot path because N was tiny. Correcting the lexicon did not merely improve extraction—it changed the computational regime of the pipeline and surfaced latent engineering debt. Lexicon quality is load-bearing in the most literal sense.

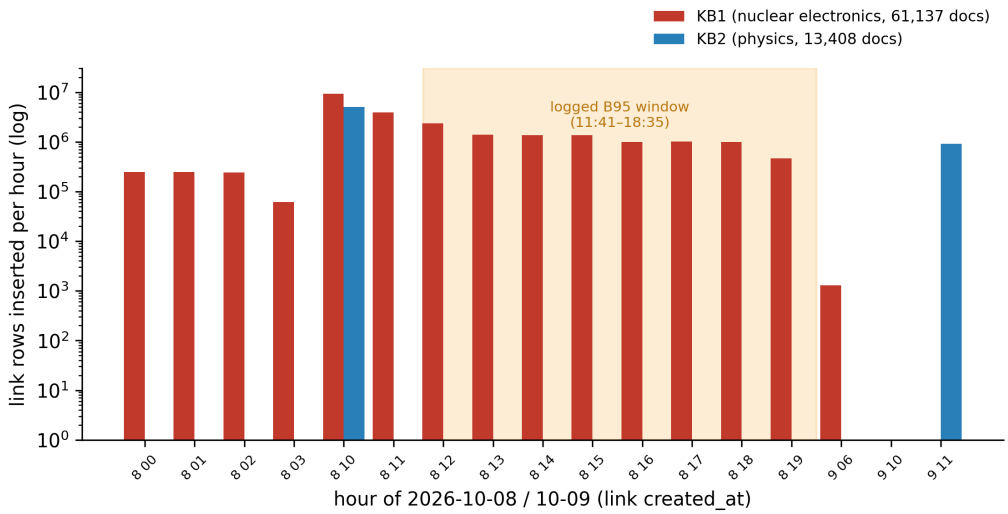


Figure 3: Rebuild telemetry from the link registry's own timestamps (hourly `created_at` counts). KB2's entire link layer lands in the 10:00 hour (5.0M rows); KB1 drains across the day at up to 9.2M rows/hour. The shaded band is the *logged* B95 window (11:41–18:35); inserts continue through it, which forces the timestamp correction reported in the text.

The rebuild also supplied its own fifth pathology, and revisiting it with the registry's own timestamps forced a correction to our incident log that we report in full. The original record: a single-statement link writer shipped with a parameter-order bug (the SQL's first placeholder expected the knowledge-base id; the call bound a term form there), jobs completed green for hours, and per-KB link counts read zero growth until a row-count reconciliation caught it. What a minimal reproduction on the production engine (MySQL 8.4, `STRICT_TRANS_TABLES`, faithful DDL) now establishes is sharper than "rows were swallowed": under `INSERT IGNORE`, the NULL produced by `UNHEX('alloy')` is **replaced by the implicit default 0x00×16 and the row is inserted**—into a ghost bucket invisible to every per-KB query and monitor (warnings 1411 and 1048 fire; nothing fails)—and a later correct re-insert for the same (mention, entity) pair is **silently skipped by the unique key**, so the orphan actively poisons the repair unless deleted first. Production carries zero 0x00 rows today, so the recovery path included that cleanup. A second correction concerns duration: the operations log recorded "seven hours of zero growth", but the surviving rows' own `created_at` histogram (Figure 3) shows KB1 inserting

at 1.0–2.3M rows/hour continuously through that nominal window and KB2's entire link layer landing in a single hour—the zero-growth reading cannot be reconciled with the timestamps and most plausibly reflects a monitoring view scoped to the buggy writer's partition rather than the whole table. We keep both the original log entry and its correction in the record: incident narratives written from job dashboards are themselves unverified claims; row-level forensics is the only authoritative account.

Rebuild final state. All four stages drained across both knowledge bases: mentions, links, relations, graph projection. The repaired entity layer, measured: **KB1 1,871 entities carrying 25,097,704 links (from 575 / ∼19.2M collapsed), 44,043 relations with 1.56M evidence rows (from effectively zero); KB2 2,730 / 5,009,755 / 12,774.** Spot-checked canonical entities (germanium detector, hpge detector, neutron, preamplifier, scintillation detector) all project correctly into the graph store. Retrieval on KB1 answers in 0.6–1.3 s warm with all channels active.

6. Before/after

Figure 4 compares entity cardinalities across the two main and two auxiliary repaired libraries.

Entity layer. The headline repair, with snapshot provenance stated (collapsed-era link metrics from the 2026-10-06 cold backup, a mid-stream snapshot two days before discovery; terminal collapsed totals from the discovery-day count; repaired metrics 2026-10-10):

Concepts-per-document replaces the raw entity/document ratio as the cardinality-family metric: its numerator scales with documents, it is readable by domain experts, and the coverage column exposes the collapse's most user-visible symptom—half the library had *no* entity annotations at all. The concentration drop 0.419→0.345 is honest about its own limits: the pre-repair figure is a mid-stream snapshot (4.81M links; the terminal state was 19.2M on the same entities, which can only push concentration toward the type-word head), and part of the drop comes from the entity population growing—the name audit below remains the sharper discriminator. Absolute link counts *rise* after repair in both libraries (KB1 ∼19.2M→25.1M; KB2 2.23M→5.0M, same-snapshot backup on both sides): the repaired lexicons carry thousands of real terms that match more often. Count was never the quality signal—distribution (concentration) and names (top-entity audit) are.

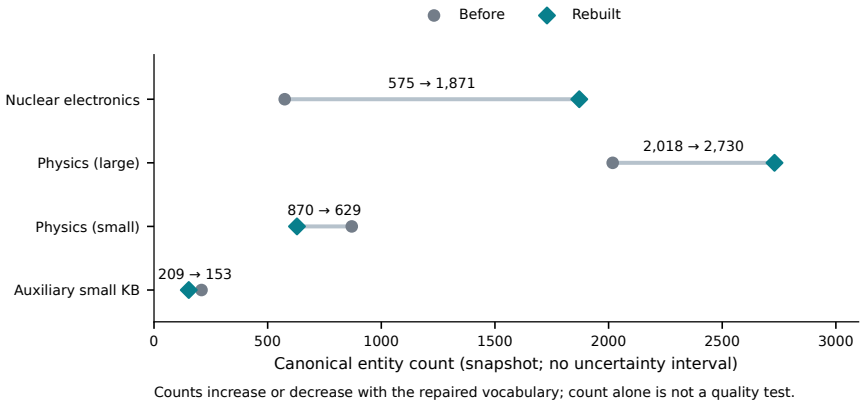


Figure 4: Entity counts before and after vocabulary repair. Gray circles are pre-repair snapshots; teal diamonds are rebuilt snapshots. Nuclear electronics and the large physics library expand, whereas the small physics library (D2B4AD67) and auxiliary library (F54CE546) contract. These are complete registry counts, not sample estimates, so no uncertainty intervals are shown. Increasing cardinality alone cannot establish improved link quality.

Name quality, honestly. The top-linked entities after repair are `be` (2.26M links), `alpha`, `gamma`, `beta`, `method`—generic words and short ambiguous symbols still dominate the link head. The corpus-attestation gate can only guarantee that a term *occurs*; it cannot distinguish a high-frequency common word from a domain term. The `be` case is instructive because it is not even a ranking problem but a normalization one: the element symbol *Be* and the English auxiliary verb *be* are distinct in the source text, but K1's scan normalizes case (ASCII terms match case-insensitively with word boundaries), so every auxiliary "be" in 7.2M chunks links to the beryllium entity. A case-sensitive allowlist for short chemical symbols plus an English stop-word exclusion at admission would remove the bulk of this mass; `alpha`/`gamma`/`beta` are harder—genuine nuclear-spectrum symbols that also serve as generic Greek-letter variables—and need context-type constraints, not stop-word lists. Genuine terminology (hpge detector, neutron, preamplifier, …) is present and correctly linked and projected—but the repair relocated the lexicon-quality problem from "no terminology at all" to "terminology plus a generic tail." We consider this the honest headline of the incident: repairing a collapsed lexicon is one battle; lexicon hygiene is a permanent operational discipline (Section 7's monitors exist precisely for this).

Table 3: Before/after, entity layer, with snapshot provenance.

Metric (KB1)	Collapsed	Repaired
canonical entities	575 (disc.) / 503 linked (10-06)	**1,871**
entity links	∼19.2M (disc.) / 4.81M (10-06)	**25,097,704**
typed relations	∼0 effective	**44,043** + 1.56M evid. rows
top-10 link concentration	0.419 (10-06)	**0.345**
concepts / document (all-doc basis)	15.7 (51.5% covered)	**37.8** (98.5% covered)
KB2: concepts / document	29.5 (91% covered)	**61.4** (91% covered)

Retrieval trajectory, with a channel-level attribution bound. On the scoped evaluation slices (same runner, same corpus scope) the system moved from the collapsed-era run (2026-09-24) to the rebuilt state (2026-10-09): associative queries cov@10 0.533→**0.599** (+6.6pt, n=48), single-intent queries 0.603→**0.675** (+7.3pt, n=39) with MRR −2.0/+5.0pt respectively. The two-week window also carried retrieval-configuration evolution (document-fusion re-ranking, bilingual alias expansion, kNN search-depth tuning), so the trajectory measures system evolution *including* the lexicon repair. To bound the attribution we ran a single-variable channel ablation under the identical post-rebuild configuration (in-process planner, same corpus scope, same questions): a *no-entity* arm with every entity-dependent channel disabled (co-occurrence, graph, entity-vector, PPR) against the full configuration. Result: **@10 any-hit and document coverage are unchanged on both families** (nea 0.599/0.938, ner 0.670/0.923); the only movement anywhere is ner MRR +0.0006 (CI [0, 0.0018], ns) and nea cov@20 −1.0pt, 95% CI [−3.1, 0.0]—while p50 latency falls from 1.46 s to 0.52 s. The ablation bounds these channels' marginal contribution in this configuration, not the semantic assets' value at large. The co-occurrence channel fired on only 6/48 association queries (its adaptive gate suppresses it when base retrieval already serves co-mention documents) and changed no outcome.

Figure 6 separates the scoped zero-effect finding from the natural-phrasing liveness control.

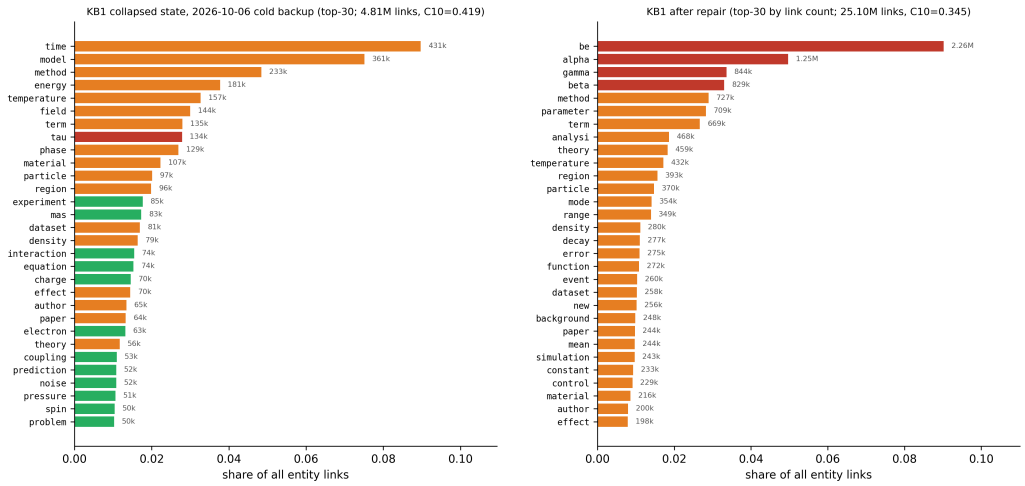


Figure 5: Top-30 entities by link share, KB1, before (2026-10-06 cold backup; $C_{10} = 0.419$) and after repair ($C_{10} = 0.345$). Categories follow a fixed three-way rubric (domain term / generic academic word / short ambiguous symbol), assigned by the authors per entity; the rubric and per-entity assignments ship with the artifact. The head is dominated by generic vocabulary in both states; repair diversifies the mid-tail (visible as the shorter bars) but does not dethrone it.

Channel-liveness check (the ablation's alternative explanation, closed). A null @10 result admits a defensible alternative reading—that the entity channels are not wired, or that their gates never open—so we ran a forced-trigger check on the 99-question natural-phrasing set (template words banned), the distribution the co-occurrence gate was designed for: there the channel fired on **54/99 queries** (versus 6/48 on the scoped template set), killing it changed the top-10 on 26 queries, and the paired gold-hit comparison reads **20 losses / 0 wins / 34 ties** for the off-arm—removing the channel never helps and hurts on a fifth of the set. The channel is alive, the gate opens exactly where designed, and the scoped zero-effect is the gate closing because base retrieval already serves those co-mention documents—not a dead wire.

Run-to-run noise floor. Two same-configuration scoped runs hours apart (the refresh run and the ablation baseline) differ on exactly one of 39 single-intent queries (cov@10 0.675 vs 0.670)—a measured 0.5pt noise floor from nondeterministic channel timing that bounds every within-window comparison in this paper.

Alias-expansion ablation. The one entity-adjacent mechanism that

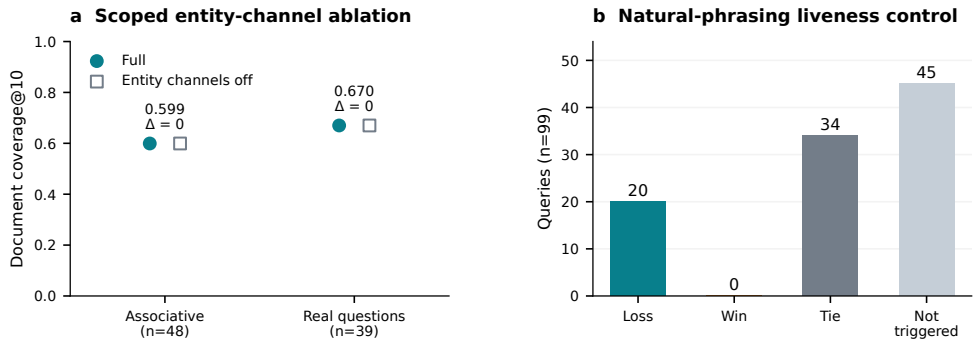


Figure 6: Distribution-dependent attribution of entity-channel effects. (a) Under the identical post-rebuild configuration, disabling all entity-dependent channels leaves cov@10 unchanged on the scoped associative ($n = 48$) and real-question ($n = 39$) sets. Points are observed set averages; no marginal confidence intervals were archived for these levels. (b) On the natural-phrasing liveness set ($n = 99$), co-occurrence fires on 54 queries: its removal causes 20 gold-hit losses, zero wins and 34 ties. The other 45 queries are shown explicitly as not triggered. The liveness experiment has a different question distribution and is not a before/after estimate of the lexicon repair's causal effect.

plausibly carried the trajectory—bilingual alias expansion feeding BM25—was ablated on/off under the final configuration: association cov@10 +1.5pt (CI [−3.9, +0.6], ns) and single-intent +0.1pt, at a p50 latency cost of 1.6→3.5 s. On the benchmark sets the retrieval-window trajectory is therefore not attributable to the lexicon through this mechanism either; where alias expansion pays is query distributions rich in dictionary surfaces (the natural-phrasing set), which the benchmarks under-represent.

The attribution this forces is precise and we state it against interest: the +6.6/+7.3pt trajectory is **not** carried by entity channels casting fusion votes. The entity layer's retrieval-facing value on these benchmark families runs through the lexical channel's bilingual alias expansion (entity names as BM25 terms—a mechanism that needs a real terminology to expand, and which the collapsed lexicon could not feed) and through structural surfaces (reranker context, knowledge-chain evidence for generation) documented in the companion system manuscript [7]; at the fusion layer itself, entity channels are currently a latency cost with zero measured recall benefit on these

sets. The lexicon repair was necessary for every entity-consuming surface; the scoped trajectory number alone cannot be credited to it.

Link precision audit (LLM-assisted). Distributional monitors say nothing about whether individual links are *correct*. We audited a seeded, reproducible uniform sample of 100 links from the rebuilt KB1 registry (`ORDER BY RAND(20261009)`): for each link the mention span was programmatically verified against the chunk text (span integrity 100/100—offsets are exact), and an LLM judge (Qwen3.6-35B-A3B, temperature 0) received the ±250-character context with the span marked and answered a strict two-part rubric: (i) does the span denote the same concept as the entity's canonical name (different sense, different concept, or non-term mention ⇒ invalid; debatable ⇒ invalid + borderline); (ii) is the canonical name a domain-specific technical concept or a generic academic word. Results: **overall precision 0.85** (Wilson 95% CI [0.77, 0.91]); the domain stratum is **25/25 = 1.00** (CI [0.87, 1.00]) and the generic stratum 60/75 = **0.80** (CI [0.70, 0.88]). Every adjudication, span, and reason string is released. The reading is the name-quality headline quantified: in this 25-link domain subsample no erroneous link was observed, and the residual error mass sits almost entirely in the generic tail—but "no errors observed at n=25" is not "error-free": the larger stratified audit's domain stratum reads 142/149, not 149/149. The audit is LLM-assisted rather than human-adjudicated—a limitation we keep visible (Section 9)—but its stratum split is exactly what the monitor design predicts, and the programmatic span check removes the cheapest class of audit noise.

Figure 7 displays the audit's uncertainty and stratum sizes.

Cross-model check (the judge-independence concern, addressed). The audit judge and the lexicon classifier are the same model family—a self-grading risk we take seriously. We re-adjudicated the identical 100 links with a second judge from a different model, checkpoint, and provider (Qwen3.8-Flash via an external gateway; temperature 0, identical rubric). Both judges are Qwen-series—vendor-level correlation is not excluded, and we label this a cross-model check, not an independent family confirmation. The second judge failed to emit parseable verdicts on 45/100—a flash-tier instruction-following failure we report as-is, not hide. The agreement figure therefore applies only to the 55-link parseable subset and cannot be read as confirmation of the full 100; on that subset, verdict agreement with the primary judge is **53/55 = 96.4%** (50 both-valid, 3 both-invalid, 2 discordant—both discordances on the borderline generic word

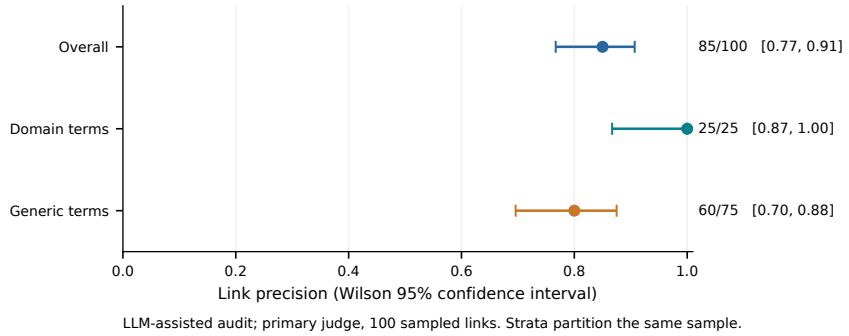


Figure 7: Link precision in the primary LLM-assisted audit of 100 sampled KB1 links. Points show valid-link fractions and horizontal intervals are Wilson 95% confidence intervals. The overall sample (85/100) comprises domain-term (25/25) and generic-term (60/75) strata; the three rows are not independent samples. Domain specificity separates the observed error mass, but the domain stratum is small and this audit is not human adjudication. Every sampled link is retained.

"term"), and the domain stratum agrees **14/14**. Where an independent model opined, it confirmed the primary adjudication; but vendor-level correlation is *not* confined to the unparseable subset—the 55 parsed links, including the domain stratum (n=25 primary, 14 cross-parsed), carry the same family risk. A human-adjudicated stratified sample remains the decisive control and is listed as such in Section 9.

Stratified 300-link audit (third review round). A uniform sample reads the head; the failure modes live in the tail. We re-audited with a stratified design: entities drawn uniformly within four strata (top-10 by link frequency; medium, ranks 11–100; tail, ranks 101+; short symbols, canonical names of $\leq$4 letters), then one link uniformly per entity—seeded, replayable, 50/100/100/50 links, same rubric and judge. Span integrity 300/300. The judged-valid rate for this sample is **0.837** (Wilson 95% CI [0.791, 0.874]); top-10 **0.920** [0.812, 0.968]; medium **0.850** [0.767, 0.907]; tail **0.910** [0.838, 0.952]; short-symbol **0.580** [0.442, 0.706]; across the specificity axis, domain 142/149 = 0.953 and generic 109/151 = 0.722. The error mass is localized, not diffuse: the stratified design confirms the uniform reading at the head and pins the failure precisely on the short-ambiguous-symbol class—the more informative finding of the audit, and the motivation for the governance pricing below (not yet a demonstration of its post-treatment validity). Two scope

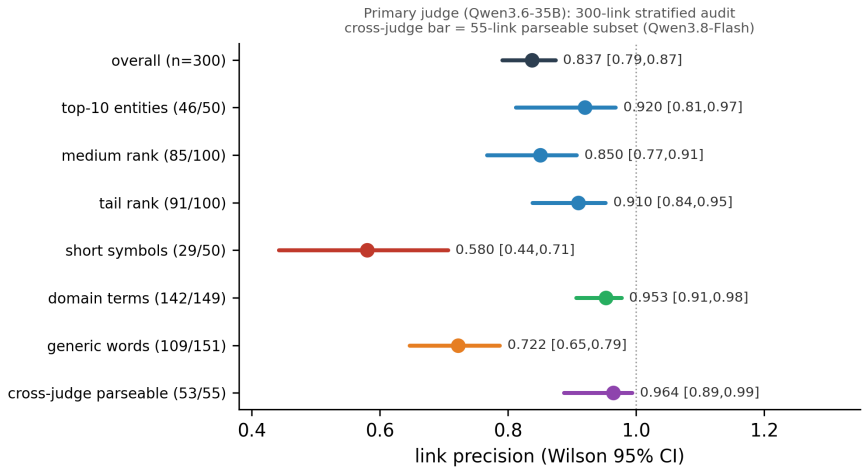


Figure 8: Link-precision with Wilson 95% CIs: 300-link stratified sample (primary judge); cross-judge bar applies to the 55-link parseable subset of the earlier uniform sample (45/100 unparseable); both judges are Qwen-series.

notes: the 50/100/100/50 quotas target risk rather than production link frequencies, so the 0.837 figure is a stratum-mixed rate, not an unweighted production-wide link precision; and the 300 links come from 187 distinct entities (the head stratum's 50 links from just six; the short and head strata share "term"), so the intervals treat links, not concepts, as exchangeable.

The lexicon variable, isolated. The repair bundled lexicon, teardown, and fleet changes; this experiment isolates the lexicon alone. Design: a semantic replica of the K1 scan-and-link semantics (CJK substring match; ASCII word-boundary + optional plural, case-folded) and the K2 fold (normalized form + type), with **only the dictionary swapped**: v2.0 as committed at library creation (182 scan terms), v2.5 (mid-evolution, 2,932), v2.1 (the repair, 3,733), plus two v2.1 governance variants (case-sensitive matching for short ASCII symbols; an English stop-word block at admission). Corpus frame: **ten seeded, uniformly random resamples of 2,000 chunks each** from the library's 5.83M visible chunks—independent draws from a *single* corpus, i.e. a sampling-stability check within one library, not cross-domain evidence—cross-sample disjoint, deduplicated on redraw. The replica copies the scanner's matching and folding rules from the same source reading (CJK substring; ASCII word-boundary with optional plural, case-folded; fold key = normalized form + type); it is not instrumented against the live service

Table 4: Isolated lexicon comparison across ten independent corpus resamples (fixed code, fixed corpus, only the dictionary differs; mean±sd). Conc/doc is the mean number of distinct concepts per *matched* document (documents with ≥1 scan hit under that arm's dictionary); the arms' denominators differ by construction (text).

Arm	Links	Ents	C_{10}	Conc/doc	Gen.-head
v2.0 (collapse-era)	$2{,}969 \pm 140$	109 ± 5	$0.513 \pm .016$	$1.85 \pm .05$	$0.146 \pm .012$
v2.5 (mid-evolution)	$4{,}823 \pm 147$	421 ± 25	$0.322 \pm .009$	$2.51 \pm .05$	$0.090 \pm .006$
v2.1 (repair)	$10{,}042 \pm 251$	692 ± 27	$0.300 \pm .008$	$3.89 \pm .07$	$0.058 \pm .004$
v2.1 + case-sens. short	$8{,}758 \pm 228$	665 ± 28	$0.265 \pm .012$	$3.54 \pm .07$	$0.066 \pm .004$
v2.1 + stop-word	$9{,}226 \pm 251$	691 ± 27	$0.252 \pm .011$	$3.67 \pm .07$	$0.063 \pm .004$

path, so absolute levels may drift—the experiment's force is the within-frame arm contrast. An earlier single run had sampled an ordered prefix—a bias we replace rather than defend; all metrics below are deterministic, reported as mean±sd across the ten resamples. Concepts-per-document is averaged over *matched* documents only (those with ≥1 scan hit under the arm's dictionary: mean 977 matched documents per sample under v2.0, 1,579 under v2.1), so it is not an all-sampled-document mean, and the two arms' denominators are different sets.

Three findings. **(1) The collapse signature is reproduced by the lexicon alone**: 109±5 entities at concentration 0.513±0.016 and 1.85±0.05 concepts per *matched* document from the v2.0 dictionary—same code, same corpus, no other variable. **(2) The repair effect is isolated**: v2.0→v2.1 multiplies link-bearing entities ×6.3 and concepts per matched document ×2.1 (the denominator grows too: mean link-bearing documents per 2,000-chunk sample rise 977→1,579), while concentration decreases 0.513→0.300 and generic-head mass falls 0.146→0.058; the production collapsed concentration (C_{10}=0.419 at the 10-06 backup) sits between the v2.0 (0.513) and v2.5 (0.322) counterfactuals, consistent with the operational dictionary having evolved mid-enrichment after the bulk of extraction had run. **(3) The "be" class is treatable at known cost**: case-sensitive short-symbol matching removes the single largest entity and takes concentration to 0.265±0.012, at the price of a wider case-fold recall loss (links −12.8%); a stop-word block is gentler (−8.1% links, C_{10} 0.252±0.011). Judged precision is near-tautological for dictionary self-matches (the entity name *is* the surface form), so the structural metrics lead and precision is a bounded secondary—the same caveat the link audit carries. These deltas

measure matching *output*, not true recall: the removed mass mixes errors with correct terms, and the 0.580 pre-intervention judged-valid rate on the short-ambiguous-symbol class motivates the treatment without yet demonstrating post-treatment validity—removed-vs-retained link verification is the open evaluation.

Why 692 entities here vs 1,871 in production? The isolated frame counts only entities induced by scanning 2,000 chunks (0.034% of the library) under replica folding: rare terms need corpus mass to be attested, and the production layer additionally carries entities from the full corpus and historical merging. The causal claim rests on within-frame comparisons, never on this cross-frame count. Methodological caveats: semantic-replica scanner, 2,000-chunk samples (ten resamples bound the variance but not the coverage of a 5.83M-chunk library), and committed-snapshot arms rather than independently regenerated component ablations (the provenance decomposition in Section 7 approximates that axis lexically).

Provenance decomposition (component contribution, approximated lexically). The v2.1 file carries no per-term source tags, so we decompose its isolated link mass by which earlier committed lexicon first contained each matched term: v2.0-inherited terms contribute 22.0% of links (70 entities), mid-evolution terms 16.0% (232), and terms first introduced by the three-source repair **61.9% of links (384 of 686 entities)**—the repair's additions carry three-fifths of the rebuilt link mass (shares stable against the retired ordered-prefix frame, which read 21.4/15.3/63.2). A true component ablation (independently regenerating single-source dictionaries) remains future work; this set-algebra attribution is its lexical lower bound. A scope note on dictionary size: the arms vary the dictionary *as a whole* (v2.0 and v2.1 differ by $\sim$20$\times$ in scan terms), so the contrast identifies the dictionary change, not its decomposition into term count, term-selection quality, and per-source method; a size-matched control (an equal-size frequency-candidate dictionary) and blind old/new matched-context semantic judging are the priority follow-ups.

7. Discussion

Observability for lexicon quality. Four cheap monitors would have caught both incidents at first symptom, in the spirit of production data-validation systems [4]:

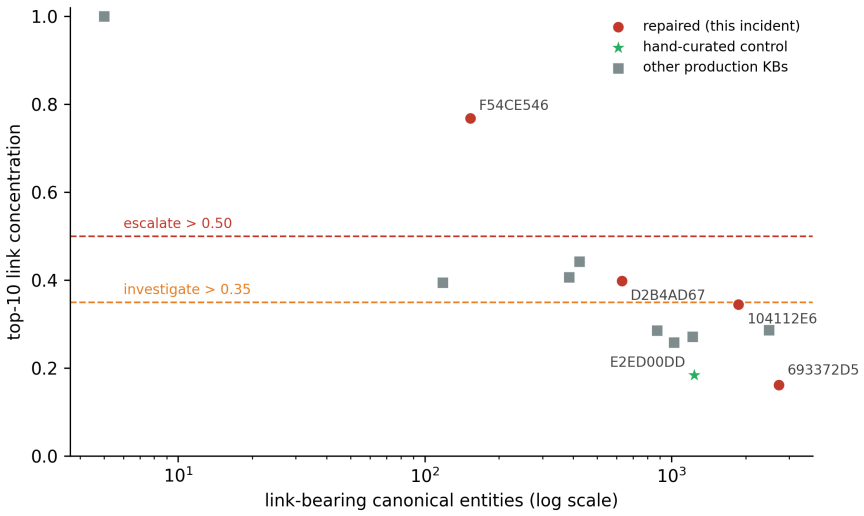


Figure 9: Top-10 link concentration vs. link-bearing entities across the 13 production libraries (log x-axis). Red = libraries rebuilt in this incident; green star = hand-curated reference case; gray = remaining production libraries. The bands are the proposed post-hoc alarm thresholds; the smallest libraries sit high partly by the Zipfian floor effect discussed in the text.

(a) *Top-10 link concentration (primary).* The fraction of all entity links held by the ten most-linked entities—a concentration diagnostic invariant to uniform rescaling of link counts. That specific invariance is what raw cardinality lacks; it is not a license to read the value as a corpus-size- or discipline-independent quality scale (topic concentration moves it too). Measured across the thirteen libraries of our deployment, the metric spans 0.16–0.77 among libraries with $\geq$100 entities (a five-entity library reads 1.000—the $\min(10, N)/N$ floor bound, not degradation; $N \leq 10$ forces C_{10}=1, and the metric also responds to topic concentration as well as to degradation): the healthy and repaired large libraries run at 0.16–0.19 (rebuilt physics KB2 0.162, hydrogeology control 0.184); the rebuilt nuclear library sits at 0.345 with its generic tail; and the two small treated libraries **currently** read 0.398 (1,324 docs, 629 entities) and 0.768 (119 docs, 153 entities)—values we report as they are, not smoothed: the smallest library also illustrates the metric's floor effect (with $O(10^2)$ entities, ten of them can hold most links under an ordinary Zipfian distribution alone), so the

band should be read jointly with library size. The collapsed state itself is now measured, not gestured at: recovering the pre-incident cold database backup (2026-10-06, two days before discovery, with the collapsed enrichment still writing—4.8M links on KB1 at that moment versus 19.2M at discovery) gives **0.419** for collapsed KB1 (top-10 entities: *time, model, method, energy, temperature, field, term, tau, phase, material*—431k links on "time" alone) and **0.290** for collapsed KB2. The repair moves KB1 0.419→0.345 and KB2 0.290→0.162—direction correct, and the residual gap is the generic tail. Note also that collapsed KB2's 0.290 sits *below* the 0.35 investigation band: a concentration threshold alone does not catch every collapse—KB2's is exposed by the name-audit and coverage legs, which is why the monitors deploy as a set. One honesty note: the metric partly improves simply because the entity population grows (503→1,871 link-bearing entities); the name audit (c) remains the sharper discriminator of the collapse itself. We propose a band alarm (>0.35 investigation, >0.5 escalation) with an explicit caveat: these are *candidate* thresholds read post-hoc on this deployment's thirteen libraries, not on an external population—a starting point for adopters, not a validated norm. Prospective validation (false-positive/false-negative rates, detection delay, and size sensitivity) is future work, and is runnable as offline graded lexicon degradation on a frozen slice rather than re-collapsing production; libraries below a few hundred entities warrant size-stratified bands. The thirteen-library candidate calibration, summarized (the collapsed row is the pre-repair backup, not a live point):

Group	Libs	Entities	C_{10} range
Collapsed (10-06 backup)	1	503	0.419
Hand-curated control	1	1,239	0.184
Other production, ≥500 entities	4	874–2,482	0.259–0.287
Rebuilt here, large	2	1,871 / 2,726	0.162–0.345
Rebuilt here, small	2	153–629	0.398–0.768
Other production, <500 entities	4	5–424	0.395–1.000

(b) *Entity cardinality ratio (coarse secondary).* Canonical entities per ingested document, alarmed against a per-domain band (our control suggests O(10–20) for scholarly corpora; collapse ran at 0.009, a factor

2,192 below the reference case, and the ratio's sublinear numerator makes it scale-sensitive—the per-document concept counts above are the better-behaved metric). Cardinality alone is gameable in the other direction—the small-KB treatment showed category-word aliases *inflating* the count (Section 9)—which is why it backs up (a) rather than leading. The O(10–20) band itself comes from a single hand-evolved control library and is offered as illustration, not a calibrated norm.

(c) *Top-entity name audit.* The top-K entities by link count should be domain concepts; a manual or LLM spot check of that list is a five-minute quarterly control, and the link-precision rubric of Section 6 is a ready-made protocol for a deeper sample.

(d) *Target-table growth reconciliation.* Completed-job counts reconciled against downstream row growth, mandatory wherever a bulk writer is guarded by INSERT IGNORE (the ghost-bucket incident of Section 5—rows landed in an all-zero kb bucket invisible to every per-KB query while job metrics stayed green—was exposed by exactly this reconciliation).

None of the four requires ground truth. The small-library treatment (Section 9) is an illustrative within-deployment case for the monitor stack—the candidate bands were read *after* observing these thirteen libraries, so this is not out-of-sample validation: after the same repair playbook, entity counts contracted (870→629, 209→153—the cardinality signal normalizes) while the concentration monitor still flags both libraries—one above the investigation threshold ($0.398 > 0.35$), one above the escalation threshold ($0.768 > 0.5$)—correctly, since their tails remain generic—demonstrating that the two monitors separate "cardinality repaired" from "distribution still skewed", which cardinality alone cannot.

LLM extraction versus dictionary scan. The incident is not an argument for LLM-everywhere. It is an argument for honoring the input contract of cheap deterministic components: if extraction is a scan, the dictionary is the model—and it deserves the same evaluation discipline (held-out attestation, distributional checks) as a trained model. Batch-generating that dictionary with an LLM and shipping it on structural validation alone is the actual failure mode.

Schema tooling implications. The batch schema generator produced category taxonomy where a terminology was needed; the repair gave the

schema a real lexicon (3,702 terms) and folded lexicon generation into the schema lifecycle as an idempotent tool. What remains open—and what the rebuilt top-entity distribution makes plain—is that *admission* criteria (corpus attestation) are necessary but not sufficient: a lexicon also needs *ranking* criteria that demote high-frequency generic words and short ambiguous symbols (IDF-style corpus contrast, bilingual alias consistency, length/type priors). The observation that the collapsed state and the repaired state differ by a factor of 3.25 in entity cardinality—while both pass structural validation—is the strongest argument we have for treating lexicon quality as first-class schema infrastructure with its own acceptance gates, not as a one-time repair.

8. Conclusion

Three questions framed this study. *When does a structurally valid lexicon produce semantic collapse?* When its entries are category words: the scan-and-link machinery executes faithfully, and a single-variable dictionary swap reproduces the collapse signature—no pipeline bug required. *What does corpus-presence gating repair, and what does it leave?* It restores terminology and coverage (annotated documents $\sim$51.5%$\rightarrow$98.5%; link-bearing entities $\times$6.3 in the isolated frame) but admits a generic tail—presence is necessary, not sufficient, and ranking criteria are the open problem. *What can gold-free diagnostics see?* Distribution and names—the collapse signature—but not link correctness: the 0.84 risk-stratified judged-valid rate, with 0.58 on short ambiguous symbols, is a model-assisted reading, and the repair's semantic gain itself still awaits blind old/new comparison on matched contexts. The production incident supplied the motivation and the validation environment; the controlled lexicon swap and the stratified audit carry the evidence.

9. Limitations and honesty

1. **The generic tail survives the repair**: top-linked entities remain short generic symbols; admission gates without ranking gates move the problem, they do not remove it.
2. **Retrieval attribution is bounded, not eliminated**: the channel ablation of Section 6 shows the trajectory does not run through entity-channel fusion votes, but the full decomposition (lexicon old/new $\times$ retrieval configuration on/off) has not been run—it is runnable on frozen slices or isolated stacks via graded degradation, not only by re-collapsing production.

3. **Coverage**: two knowledge bases rebuilt (61,137 + 13,408 documents) and, after the two main libraries drained (2026-10-09), the two smaller affected libraries (1,324 + 119 documents) with the same playbook—where entity cardinality *contracted* (870→629, 209→153, with 916k/26k links and 4.6k new typed relations): the category-word aliases had inflated the count, so cardinality is not the quality metric, name distribution is. The bilingual alias layer showed isolated anomalies (an entity whose mentions vanished between extraction generations) that we document but do not fully explain.
4. **Incident-response cost**: six secondary incidents across the repair window (one INSERT-IGNORE ghost-bucket mis-write whose logged duration was corrected by row-level timestamps, three half-open-connection wedges, a watchdog that mistook completion for failure and deleted the freshly rebuilt graph, one missing-index class) each extended the repair; we report them as findings, not hidden costs.
5. **No human-adjudicated ground truth** for entity correctness: the link-precision audit is LLM-assisted (with programmatic span verification); the stratified 300-link sample replaces the uniform 100 but entity-uniform sampling intentionally over-weights the tail relative to link frequency, and the cross-judge is same-vendor-family. A human-adjudicated stratified sample is the natural next evaluation step.

Data availability

The lexicon-generation tool (three-source, corpus-attested), the rebuild driver, the link-precision audit driver with all 400 adjudications (the uniform 100 plus the stratified 300—spans, verdicts, reasons), the isolated-comparison and resample drivers with their sampled chunk-id manifests, the channel-ablation runner, and the observability queries (top-10 link concentration, cardinality ratio, top-name audit, row-growth reconciliation) are released in a public artifact mirror at `https://github.com/simpleqt/knowledgenexus-artifact` (no access request needed; the institutional GitLab carries the full development history). For double-blind venues an anonymized mirror will be provided with the submission. Schema lexicons (nuclear_electronics v2.1, physics v1.1) ship as versioned assets in the same repositories. The journal corpus is third-party copyrighted; entity-layer aggregate statistics and name samples

sufficient to reproduce the forensic analysis are tabulated in the released evidence pack.

Conflict of interest

The authors declare no competing interests.

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